

Nova Bank Project Report

Project: Hack-To-Night 2026 Challenge / Nova Bank
Stack: Next.js 16, React 19, Tailwind CSS, MySQL, Firebase, Nodemailer, Biome
Report output: `reports/project-report.html` and `reports/project-report.pdf`

Summary: This report documents the project functions, new features added, errors found during development, how those errors were solved, and the tools/methods used to verify the application.

1. Project Overview

Nova Bank is a digital banking web application with public landing pages, account authentication, customer dashboard features, bank transfer flows, bill payments, savings goals, e-statements, profile/settings management, notifications, and an AI banking assistant.

2. Main Project Functions

Area	Functionality	Main Routes
Public landing	Modern marketing landing page, animated hero, colorful banking carousel, footer navigation.	<code>/</code>
Authentication	Username/password login, signup, reset password page, logout, current user lookup, Google/Firebase login support.	<code>/login</code> , <code>/sign-up</code> , <code>/reset-password</code>
Dashboard	Total balance, monthly spend, category chart, recent transactions, payees, shortcuts.	<code>/dashboard</code>
Accounts	View, create, update, and manage bank accounts.	<code>/bank-accounts</code> , <code>/accounts</code>
Transfers	Send money, validate account/PIN information, save beneficiaries, transaction history.	<code>/bank-transfer</code>
Bill payments	Pay supported billers and record payment status/history.	<code>/pay-bills</code>
Smart Spend	Spending insights, budget targets, account summaries, recent bills, currency rates.	<code>/smart-spend</code>
Savings goals	Create goals, update targets, track saved amount and daily saving plan.	<code>/savings-goals</code>
E-statement	Generate account statement preview and transaction details by account number.	<code>/e-statement</code>
Profile/settings	Edit profile, avatar, email, NIC, and local UI preferences.	<code>/profile</code> , <code>/settings</code>
Notifications	Read email notification queue records, status, subject, body, and dates.	<code>/notifications</code>

AI assistant	Floating chatbot and full chatbot page for banking help.	/chatbot
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3. API Functions

API	Methods	Purpose
/api/auth/login	POST	Authenticate user with username/password.
/api/auth/logout	POST	Clear session cookie.
/api/auth/me	GET	Return current authenticated user.
/api/auth/signup	POST	Create new customer account.
/api/auth/firebase	POST	Login using Firebase/Google ID token.
/api/accounts	GET/POST/PUT/DELETE	Manage bank accounts.
/api/transfer	POST	Execute transfers.
/api/beneficiaries	GET/POST/DELETE	Manage saved payees.
/api/bill-payments	POST	Record bill payment requests.
/api/transactions	GET	Return account transaction history.
/api/statement	GET	Build statement data by account number.
/api/smart-spend	GET/POST/PUT	Insights and budget target management.
/api/savings-goals	GET/POST/PUT/DELETE	Savings goal CRUD operations.
/api/profile	GET/PUT	Profile read/update and notification queueing.
/api/notifications	GET	Read notification queue records.
/api/chatbot	POST	AI assistant request handling.
/api/health	GET	Service health check.
/api/setup	GET	Database setup/boot support.
/api/admin/system	GET	Admin-only system data.
/api/search	GET	Search endpoint.

4. New Features Added

Feature	What Was Added	Files/Areas
Modern landing page	Rebuilt landing from simple centered content into a premium hero with animated visual cards and green/cyan/amber accents.	<code>app/page.tsx</code> , <code>app/globals.css</code>
Large carousel	Converted the standalone carousel idea into a reusable React carousel using local assets and a larger 3D rotating ring.	<code>components/carousel.tsx</code>
Scroll animations	Added global scroll reveal behavior for sections, cards, forms, and landing elements.	<code>components/scroll-animations.tsx</code> , <code>app/layout.tsx</code>
Settings page	Added protected settings page with profile details, avatar selection, notification preferences, and security status.	<code>app/settings/page.tsx</code>
Notifications page	Added protected page that reads from <code>/api/notifications</code> , with loading, empty, and error states.	<code>app/notifications/page.tsx</code>
Navigation/protection	Updated sidebar and route protection for settings and notifications.	<code>components/sidebar.tsx</code> , <code>proxy.ts</code>
Footer	Added a full landing footer with product/account/support links.	<code>app/page.tsx</code>

5. Errors Found and Solutions

Error / Issue	Cause	Solution
Standalone <code>carosel</code> folder was not integrated	It contained plain HTML/CSS and referenced missing <code>assets/*</code> images with unrelated Marvel content.	Recreated the carousel as a React component using existing banking images from <code>public/</code> .
Missing protected routes for new pages	<code>/settings</code> and <code>/notifications</code> were not listed in the route guard.	Added both to <code>protectedPages</code> and middleware matcher in <code>proxy.ts</code> .
Notifications UI missing	The API existed but no user-facing page was present.	Created <code>/notifications</code> page with refresh, empty, loading, and error handling.
Settings UI missing	Profile page existed, but there was no dedicated settings page.	Created <code>/settings</code> page and reused <code>/api/profile</code> for data updates.
Biome formatting/import errors	Manual file edits created formatting and import-order differences.	Ran <code>npx biome format --write</code> on touched files and verified with <code>npm run lint</code> .

UTF-8/mojibake text such as <code>Ã©</code>	PowerShell write encoding produced incorrect display for special characters.	Rewrote affected files with UTF-8 without BOM and changed footer copy to ASCII-safe text.
TypeScript error: missing <code>CSSProperties</code>	Carousel inline custom CSS variables needed a type import.	Added <code>import type { CSSProperties } from 'react'</code> .
<code>npm</code> not valid Win32 app when starting server	Windows requires the command shim for background launch.	Used <code>npm.cmd</code> for <code>Start-Process</code> .
<code>spawn EPERM</code> during build/dev in sandbox	Windows sandbox blocked process spawning for Next workers.	Reran build/dev with scoped elevated approval. Build completed successfully.
Landing hero looked too plain/centered	Original hero was mostly text-centered with limited visual hierarchy.	Rebuilt hero with asymmetric layout, animated finance cards, scan line, grid, and gradient headline.

6. Methods and Tools Used

Tool / Method	Usage
<code>rg</code>	Fast file and text search for routes, APIs, existing components, and styles.
PowerShell	Read files, run checks, inspect process/server status, and export PDF.
<code>apply_patch</code>	Created new report and component files safely.
Biome	Formatting and static checks via <code>npm run lint</code> .
Next build	Production validation via <code>npm run build</code> .
Microsoft Edge headless print	Exported the HTML report to PDF.
Manual route review	Mapped all pages and APIs from the filesystem and <code>app/api/route.ts</code> .

7. Testing and Verification

- **Passed:** `npm run lint` using Biome.
- **Passed:** `npm run build` after approved elevated run because sandbox build hit `spawn EPERM`.
- **Passed:** Dev server responded with HTTP 200 at `http://localhost:3000`.
- **Checked:** New protected routes are included in `proxy.ts`.
- **Checked:** Landing page, carousel, settings, and notifications compile in production build.

8. Future Improvements

- Add automated UI tests for login, dashboard, transfer, settings, and notifications.
- Add persistent user preference storage for settings toggles.
- Add read/unread status and actions to notifications.
- Add real charting library for spending analytics if the project grows.
- Add accessibility audit for color contrast, keyboard focus, and reduced-motion behavior.

- Replace demo seed data with environment-specific fixtures for production readiness.

9. Final Status

The project now has a stronger public landing experience, larger banking carousel, protected settings and notifications pages, route protection updates, global animation support, and verified production build output.